

Tanish Arora

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LinkedIn

GitHub

Portfolio

EDUCATION

Vellore Institute of Technology, Chennai

B.Tech in Computer Science Engineering

Aug 2023 – Expected June 2027

CGPA: 8.79

TECHNICAL SKILLS

- **Languages:** C++, JavaScript, TypeScript, SQL, Bash/Shell
- **Frameworks & Libraries:** React, Node.js, Express, FastAPI, Tailwind, REST APIs
- **Databases & Cache:** PostgreSQL, Redis, MongoDB, SQLite
- **Data & Machine Learning:** FAISS, Scikit-learn, XGBoost, ARIMA, Pandas, NumPy
- **DevOps & Architecture:** AWS (EC2, ECS, ECR, ALB), Docker, Celery, GitHub Actions, Linux
- **Developer Tools:** Git, GitHub, VS Code, Postman, Jupyter Lab

EXPERIENCE

HCLTech

AI & GenAI Research Intern

May 2026 – Jul 2026

Remote

- Contributed to AI-driven **EEG signal analysis** projects, leveraging **Transformer architectures** and **KANs** for time-series modeling.
- Developed data preprocessing pipelines and performed feature extraction to improve model classification performance.
- Conducted experiments evaluating deep learning models for biomedical signal processing and real-world neuroscience applications.

Sahii.in

Software Developer Intern

Apr 2026 – May 2026

Remote

- Architected web apps using **React**, **Node.js**, **FastAPI**, applying SEO optimization (WebP, robots.txt) to boost Light-house scores by 40%.
- Optimized **MongoDB** indexing and query structures, reducing average backend API response latency to **15ms** and accelerating load times.
- Streamlined cloud workflows by containerizing applications with **Docker** and migrating datasets to **AWS** and **Cloudflare R2**.
- Independently resolved complex architectural challenges while collaborating effectively within a fast-paced agile team environment.

PROJECTS

Commons – Real-Time AI Moderation Gateway [Live Demo](#) | [GitHub](#)

Feb 2026

React, FastAPI, Node.js, Redis, FAISS, Celery, PostgreSQL, Sentence-Transformers

- Architected a sub-second moderation pipeline using **FastAPI** and **Node.js** for concurrent, multilingual text and voice streams.
- Engineered a **Redis** pre-filter to intercept ~70% of invalid traffic in <10ms, drastically reducing LLM inference costs.
- Integrated **FAISS** for <20ms semantic search on banned embeddings, cascading edge cases to a Groq LLM to prevent hallucinations.
- Designed an asynchronous architecture offloading audit logs to **PostgreSQL** via **Celery** workers for instant client feedback.

NeuroSense – AI-Powered Dementia Screening Platform [Live Demo](#) | [GitHub](#)

Mar 2026

React, Node.js, FastAPI, MongoDB, Docker, Scikit-learn, Librosa, Jest, Pytest

- Architected a 3-tier microservice platform (**React**, **Node.js**, **FastAPI**, **MongoDB**) merging cognitive metadata with acoustic speech features.
- Extracted speech biomarkers via **Vosk** and **Librosa**, training a **scikit-learn** model on 1,600 samples (testing on 400) for an **89.5% ROC-AUC**.
- Engineered an API gateway featuring multi-strategy **OAuth**, **RBAC**, automated PDF reporting, **Helmet** security, and rate limiting.
- Enhanced observability with **Pino** logging; validated API, ML, UI, and E2E flows using **Jest**, **Pytest**, **Vitest**, and **Playwright**.
- Automated deployments using multi-stage **Docker** builds and **GitHub Actions**, securing VPS infrastructure via a **Cloudflare Proxy Tunnel**.

Secure Plus – Privacy-Focused Document Redaction Tool [GitHub](#)

Dec 2025

Python, OpenCV, Node.js, AWS ECS, Docker, MERN Stack

- Developed a **Python** and **OpenCV** engine to automatically detect and redact sensitive PII (PAN, emails, DOB) from documents.
- Implemented irreversible **Gaussian blur masking** to secure selected regions while preserving the original document layout.
- Deployed image-processing workloads to containerized Python workers on **AWS ECS** to ensure scalable architectural performance.